

EdParth Study Platform

Next-Gen Education Engine & NTA-Standard CBT Simulation

Target Exams:	CBSE Class 9-12 JEE Main & Advanced NEET UG NDA SSC CUET
Core Framework:	React 19, TypeScript, Vite, Tailwind CSS v4
Cloud Infrastructure:	Firebase Authentication, Cloud Firestore (Real-time NoSQL), Google Analytics 4
Storage Architecture:	Rs 0 Unlimited Free Tier via Telegram Books Channel (@edparthbooks) & Google Drive
Security & DRM:	SHA-256 Auth Fallback, DRM 4-Layer Anti-Theft Guard, Anti-Inspection
Document Version:	Version 2.4 (Production Launch Ready)
Lead Architect & Dev:	RAJ KANNAUJIYA (Developer & System Architect)
Engineering Team:	TEAM PARTH
Release Date:	September 2026
Official Domain:	edparth.com GitHub: github.com/raj-parth/edparth

EXECUTIVE SUMMARY: EdParth Study is an enterprise-grade digital education ecosystem engineered to bridge tier-2 and tier-3 Indian students with premium competitive test prep. It features an exact NTA-standard Computer Based Test (CBT) engine, an interactive mascot ('PARTH') with privacy eye-covering modes, real-time Firebase synchronization across devices, and a revolutionary Rs 0 unlimited storage architecture.

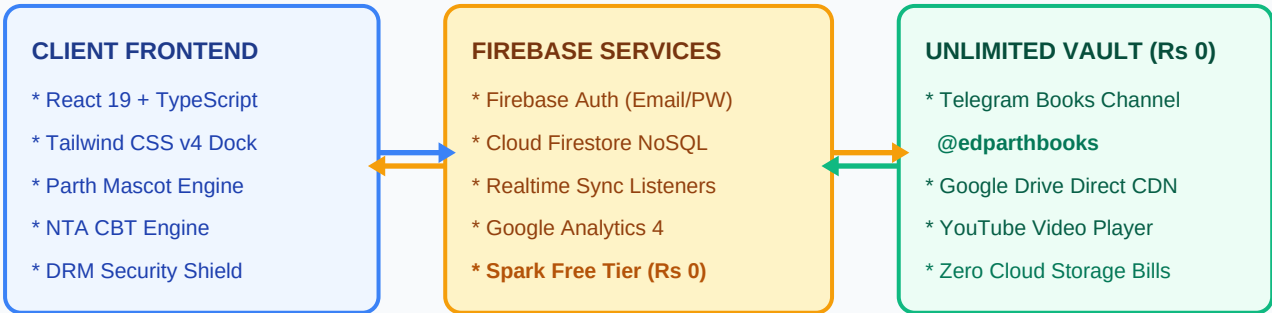
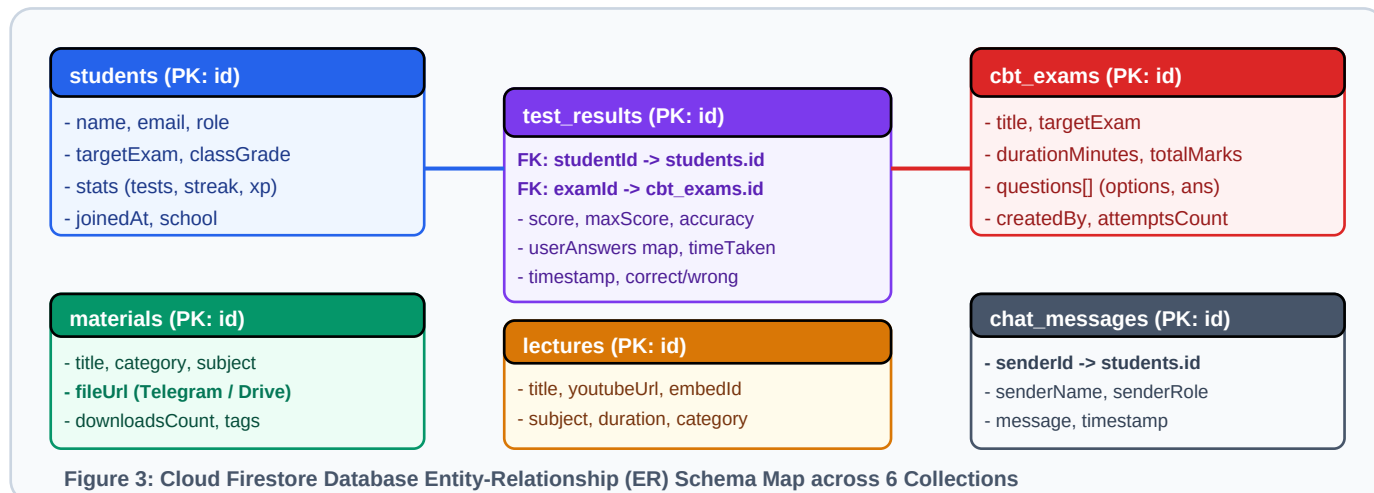


Figure 1: Full-Stack Cloud & Client Topology with Zero-Cost Storage Architecture

school	string	Optional	Name of the school or coaching institution the student attends.
phone	string	Optional	Student contact telephone number for OTP or broadcast alerts.
passwordHash	string (SHA-256)	Optional	Fallback SHA-256 encrypted password digest when operating in offline fallback.
stats.testsGiven	number	Yes	Total count of NTA CBT examinations completed by the student.
stats.studyHours	number	Yes	Accumulated focus study hours tracked via the Pomodoro module.
stats.streakDays	number	Yes	Consecutive daily login streak counter with automatic daily increment.
stats.avgScore	number (percentage)	Yes	Rolling average percentage accuracy across all completed CBT tests.
stats.xp	number	Yes	Gamification XP score earned via test completions and streak maintenance.
joinedAt	string (ISO-8601)	Yes	Timestamp of account creation (e.g., 2026-09-12T10:00:00.000Z).

3. Cloud Firestore Database Schema & ER Model

EdParth Study utilizes Google Cloud Firestore as its real-time NoSQL document store. The data architecture is organized into 6 primary collections with indexed relationships for rapid querying and zero lag.

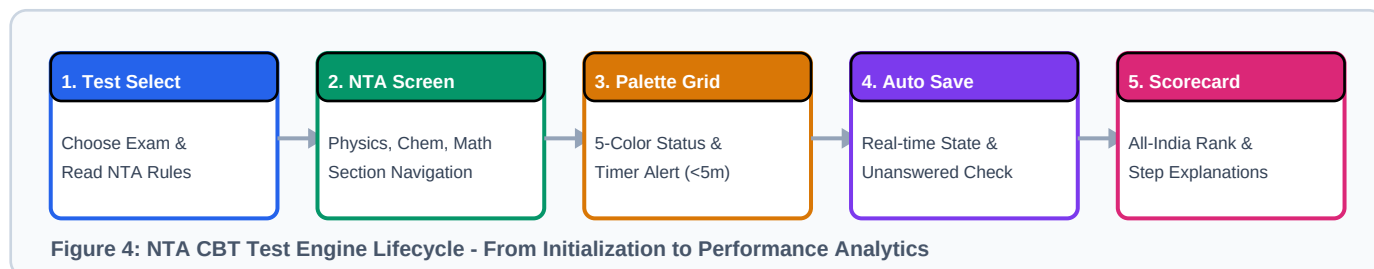


Detailed Firestore Collections Breakdown:

- students:** Stores complete user profile, role credentials, academic tags, and real-time gamification stats.
- cbt_exams:** Full question banks, section definitions (Physics, Chemistry, Maths/Biology), timer limits, +4/-1 marking formulas, and answer explanations.
- test_results:** Immutable student submission logs containing studentId, examId, score, percentile, time spent, and detailed option selection map.
- materials:** Curated reference books, handwritten notes, DPP sheets, formula cards, and direct links to Telegram/Google Drive.
- lectures:** Organized YouTube video lecture catalog with video embed IDs, subjects, tags, teacher names, and duration.
- chat_messages:** Community doubt and discussion messages with timestamps, sender role badges, and material request flags.

4. Hyperrealistic NTA-Standard CBT Examination Engine

The core USP of EdParth Study is its computer-based examination engine built to replicate the National Testing Agency (NTA) interface utilized in JEE Main/Advanced and NEET UG examinations. Students experience the genuine pressure and format before their actual exam.



NTA Question Palette State Legend:

Status Code	Visual Indicator	Exam Impact	NTA Standard Meaning
answered	Bright Green Box	Evaluated (+4 / -1)	Student has selected an option and saved their response.
not_answered	Bright Red Box	0 Marks	Student visited the question but chose not to answer.
not_visited	Silver Grey Box	0 Marks	Question has not yet been opened or viewed by the student.
marked_for_review	Deep Violet Circle	0 Marks	Marked for reconsideration; option not chosen. Will NOT be evaluated.
answered_marked_for_review	Violet with Green Dot	Evaluated (+4 / -1)	Option selected, but flagged for final review. NTA evaluates this for marks.

5. Mascot 'PARTH' & Zero-Cost Cloud Storage Strategy

A. Interactive AI Mascot 'PARTH'

To alleviate student anxiety during rigorous prep, EdParth integrates an intelligent SVG mascot named PARTH. Equipped with inverse kinematic trigonometry, PARTH smoothly tracks the student's cursor and gestures across screen coordinates. Crucially, when the student clicks into any password input field, PARTH activates **Smart Password Privacy Mode**, physically bringing his robotic hands up to cover his eyes!

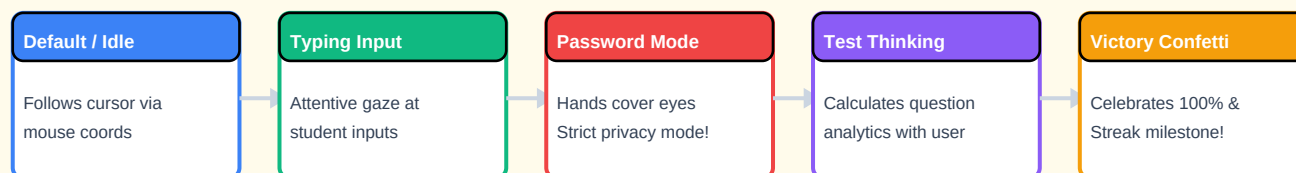


Figure 5: Interactive Mascot 'PARTH' - 5 Behavioral Reactive States & Privacy Eye-Covering Mode

B. Revolutionary Zero-Cost (Rs 0) Storage Architecture

Standard cloud file storage (e.g., Firebase Storage, AWS S3) imposes strict credit card requirements and per-gigabyte bandwidth fees. To ensure EdParth operates perpetually at **Rs 0 cost with no credit card billing risk**, the platform utilizes a hybrid storage architecture:

- 1. Telegram Channel (@edparthbooks)**: Large PDF books, test series question banks, and notes are hosted directly on the official Telegram channel, allowing free, unlimited file hosting up to 2GB per file.
- 2. Google Drive Direct CDN**: Teachers and administrators can paste Google Drive shareable links which the frontend automatically transforms into high-speed direct download/stream URLs.
- 3. Firebase Spark Cloud Firestore**: Firestore stores only lightweight metadata (file names, sizes, categories, and direct links), keeping the database well within the generous 1GB free daily quota.

6. DRM Anti-Piracy Security Shield

Educational content creators frequently face unauthorized piracy of proprietary test papers and notes. EdParth Study incorporates a multi-tiered Digital Rights Management (DRM) guard active across the entire client application:

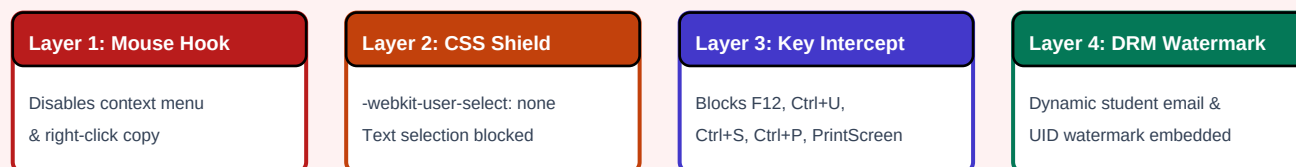


Figure 6: 4-Layer Enterprise DRM & Anti-Piracy Guard Protecting Platform Content

Anti-Theft Enforcement Rules:

- * **Mouse Shield**: contextmenu event is intercepted and cancelled, preventing right-click 'Save Image As' or 'Inspect'.
- * **Selection Guard**: Global CSS rule user-select: none prevents highlighting and copying question text.
- * **Keyboard Interceptor**: Blocks key combinations for DevTools (F12, Ctrl+Shift+I, Ctrl+Shift+J), Source inspection (Ctrl+U), Printing (Ctrl+P), and Saving (Ctrl+S).
- * **Dynamic Watermarking**: Test screens inject a subtle, semi-transparent overlay displaying the logged-in student's email and timestamp to deter mobile camera captures.

7. Complete Platform Modules Breakdown

Module	Key Features & Capabilities	Primary User
1. Mascot Guide	Cursor-tracking eyes, eye-covering password mode, reactive celebration confetti on test submission.	All Students
2. NTA CBT Engine	Section tabs, live 5-minute countdown alert, question palette grid, clear response, instant percentile scorecard.	JEE/NEET Students
3. PDF-to-CBT Converter	Instant document parsing of question banks into full digital CBT exams with diagram attachments.	Admins & Teachers
4. Study Vault	Categorized repository for Class 9-12 NCERT, JEE/NEET modules, DPPs, and Telegram books (@edparthbooks).	All Students
5. Video Lecture Hub	Distraction-free YouTube player organized by subject, class, and educator channels with no side recommendations.	All Students
6. AI Doubt Solver	Instant step-by-step solutions for complex physics derivations, chemical equations, and calculus proofs.	All Students
7. Community Chat	Real-time peer discussion forum with verified educator badges and study material request desk.	Students & Mentors
8. Gen-Z Widgets	Pomodoro focus clock (25m/5m cycles), 3D flipping formula flashcards, daily streak counter & badges.	All Students
9. Master Admin Hub	Secure dashboard (/admin) to monitor registered students, create tests, upload notes, and view analytics.	Platform Admins
10. Mobile Navigation Dock	App-like floating glassmorphic bottom bar optimized for thumb-friendly mobile navigation on phones.	Mobile Users
11. DRM Security Guard	4-layer anti-piracy shield preventing right-click, text copy, DevTools access, and screen recording.	Security System

8. Administrator Operations & Deployment Guide

A. Secret Admin Portal Access

For maximum security, there is no public admin button. Administrators can access the master console using any of these URLs: `/admin`, `?admin=true`, or `?portal=admin`. The system validates administrator role permissions in Firestore before granting access to student management or test creation tools.

B. Production Deployment Checklist

1. Push latest source code to GitHub: `git push origin main`
2. Connect GitHub repository to Vercel (vercel.com) or Netlify.
3. Configure Build Settings: Framework: Vite | Build Command: `npm run build` | Output Directory: `dist`
4. In Firebase Console, ensure Cloud Firestore and Authentication are enabled in production mode.
5. Verify Google Analytics 4 stream is receiving events with Measurement ID: `G-V7BWEXCZ7G`.
6. Test live site across mobile viewports to verify floating bottom navigation and responsive NTA CBT test palette.

ARCHITECTURAL SIGN-OFF & CREDITS
Lead Architect & Developer: RAJ KANNAUJIYA Core Engineering Team: TEAM PARTH Platform: EdParth Study (edparth.com) Status: Production Ready & Verified * 0 Build Errors